

TPA Correction in Second-Moment Beam Width Analysis

Simulation study for ideal Gaussian beams

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1 Background

Silicon-based cameras used at wavelengths above approximately 1100 nm (the silicon single-photon bandgap) detect via two-photon absorption (TPA). The detected signal is proportional to the square of the local intensity:

$$S(x, y) = I(x, y)^2 + B + \varepsilon(x, y), \quad (1)$$

where B is a uniform background (dark current) and ε is additive noise. To apply the ISO 11146 second-moment method, which requires the linear intensity distribution $I(x, y)$, the BEAMPRO-FILER application applies a sign-preserving square-root correction after background subtraction:

$$\hat{I}(x, y) = \text{sign}(S - B) \cdot \sqrt{|S - B|}. \quad (2)$$

This is the physically motivated approach and is the order recommended by ISO/TR 11146-3 §3.1 (subtract background first, then transform).

An alternative, simpler approach sometimes used in practice is to analyse the raw TPA signal directly (without the $\sqrt{\cdot}$ transform) and then multiply the resulting beam width by $\sqrt{2}$. For a perfect noiseless Gaussian beam this is exact:

$$S(x) \propto \exp\left(-\frac{4x^2}{w^2}\right) = \exp\left(-\frac{2x^2}{(w/\sqrt{2})^2}\right), \quad (3)$$

so $\sigma_S = \sigma_I/\sqrt{2}$, and multiplying by $\sqrt{2}$ recovers σ_I exactly.

2 Simulation

Both approaches were evaluated using Monte Carlo simulation on a TPA camera model ($S = I^2 + B + \varepsilon$) with the following fixed parameters: beam/sensor ratio $w_0/r_{\text{sensor}} = 0.20$, background fraction $f_B = B/S_{\text{peak}} = 5\%$, ISO statistical background subtraction (§3.4.2), and $N = 50$ noise realisations per condition. The camera is 8-bit ($\text{ADC}_{\text{max}} = 255$), pixel pitch $5.0\ \mu\text{m}$, sensor 256×256 pixels.

Two beam parameters were swept independently to test the robustness of each approach:

- M^2 from 1.0 to 5.0 (with $M_y^2 = 1.2 M_x^2$).
- Astigmatic separation $\Delta z_0 = z_{0y} - z_{0x}$ from 0 to $5 z_R$.

In all cases the measurement is taken at the x-waist position ($z = z_{0x}$). The errors are expressed relative to the true σ_x of the linear intensity profile.

3 Results

3.1 Comparison at fixed conditions

Figure 1 shows the relative error of both approaches as a function of beam/sensor ratio and background fraction.

The **TPA-corrected pipeline** (Eq. 2) underestimates σ_x by approximately 13–15% across all conditions tested. This systematic bias is not caused by an error in the correction formula itself but by the interaction between the $\sqrt{\cdot}$ transform and the background subtraction: after subtracting the estimated background $\hat{B}$, the residual image contains small negative values (noise below the background level). The sign-preserving sqrt maps these to small negative values in the linear domain. The iterative second-moment mask then integrates over a slightly different effective profile than the true $I(x, y)$, and the nonlinearity of the sqrt preferentially weights the beam centre over the wings, causing the second moments to underestimate the true width.

The **uncorrected $\times\sqrt{2}$ workaround** underestimates σ_x by only approximately 3% across the same conditions. The smaller bias arises because the TPA signal I^2 is itself a well-behaved Gaussian (for a Gaussian I), so the second-moment integration is self-consistent. The background subtraction residuals are smaller relative to the signal in the I^2 domain (since the signal-to-background ratio is higher at the beam peak), and multiplying by $\sqrt{2}$ recovers σ_I with very little residual error.

3.2 Dependence on M^2 and astigmatic separation

Figure 2 shows the errors as a function of M^2 and astigmatic separation.

Both errors are **independent of M^2** (from 1.0 to 5.0) and **independent of astigmatic separation** (from 0 to $5z_R$). The corrected pipeline gives $\approx -13\%$ and the workaround gives $\approx -3\%$ regardless of beam quality or astigmatism.

This result is not surprising in retrospect. M^2 controls the *rate of divergence* of the beam but does not alter the Gaussian shape of the intensity profile at any individual cross-section. At any fixed z-position, both an $M^2 = 1$ and an $M^2 = 5$ beam present the camera with a perfect Gaussian intensity distribution; they differ only in the numerical value of $w(z)$. Since the simulation keeps w_0/r_{sensor} fixed across the M^2 sweep, the actual beam size seen by the camera is identical in all cases, and the errors are therefore identical.

Similarly, astigmatic separation changes the ellipticity of the beam at the measurement position but not the Gaussian shape of either principal axis. The x and y second moments are separable for an axis-aligned Gaussian and each axis sees the same single-axis error independently.

4 Discussion

For an ideal Gaussian beam the uncorrected $\times\sqrt{2}$ approach is substantially more accurate than the TPA correction implemented in BEAMPROFILER: -3% vs -13% . The advantage is consistent across all tested M^2 values, background levels, and astigmatic configurations.

The -3% residual error of the workaround arises from background subtraction imperfection and noise, not from the $\sqrt{2}$ factor itself. Improving the background estimation (e.g. using a separately acquired dark frame) would reduce it further.

Limitation: non-Gaussian beams

The $\sqrt{2}$ factor is exact only for Gaussian beams. For a beam with a non-Gaussian intensity profile — such as a Hermite-Gaussian or Laguerre-Gaussian higher-order mode, a flat-top beam,

a beam with hot spots or ring structure, or any multimode superposition — the TPA signal I^2 is *not* a rescaled version of I . The cross-terms in $(I_1 + I_2 + \dots)^2$ produce a profile that differs qualitatively from I itself, and the simple $\sqrt{2}$ rescaling no longer applies.

In such cases the TPA correction (Eq. 2) should be used instead, as it recovers $I(x, y)$ pixel-by-pixel without any assumption on beam shape. The $\approx -13\%$ systematic underestimate then applies, but it is preferable to an unknown and beam-shape-dependent error that the $\sqrt{2}$ workaround would introduce.

4.1 Comparison with a linear camera

Figures 3 and 4 compare all three approaches against a linear camera analysed without any TPA correction (the standard case). The mean errors are summarised in Table 1.

Analysis approach	Mean σ_x error
Linear camera (no correction)	-3.0%
TPA uncorrected $\times \sqrt{2}$	-3.3%
TPA corrected (Eq. 2)	-14.8%

Table 1: Mean relative σ_x errors averaged over all beam/sensor ratios (10–75%) and background fractions (1–10%). ISO statistical background subtraction, 50 noise realisations per condition.

The linear camera and the TPA uncorrected $\times \sqrt{2}$ workaround give nearly identical errors (-3.0% vs -3.3%) across the full range of tested conditions. Both are essentially flat with respect to beam/sensor ratio and background fraction, indicating a consistent small bias from background subtraction imperfection rather than any systematic camera- or transform-related effect.

The TPA corrected pipeline gives -14.8% on average and shows a mild dependence on background fraction (larger background $\rightarrow$ slightly larger underestimate), consistent with the interpretation that the $\sqrt{\cdot}$ transform amplifies background residuals in the beam wings.

The conclusion is that the TPA uncorrected $\times \sqrt{2}$ workaround recovers the true Gaussian beam width with the same accuracy as a linear camera under the same conditions. The physically motivated TPA correction introduces a $\approx 12\%$ additional systematic error relative to both.

Practical recommendation

- For beams that are known or expected to be well-described by a Gaussian profile (stigmatic or simple astigmatic, M^2 not excessively large, no hot spots), the uncorrected $\times \sqrt{2}$ approach gives smaller absolute errors and is recommended. Its accuracy matches that of a linear camera.
- For beams with unknown or complex profiles, apply the TPA correction (Eq. 2) to avoid beam-shape-dependent biases. Accept the $\approx -15\%$ systematic underestimate as an upper bound on the error.
- In either case, use ISO statistical background subtraction (§3.4.2) and keep the beam waist below 25% of the sensor half-width.

5 Figures

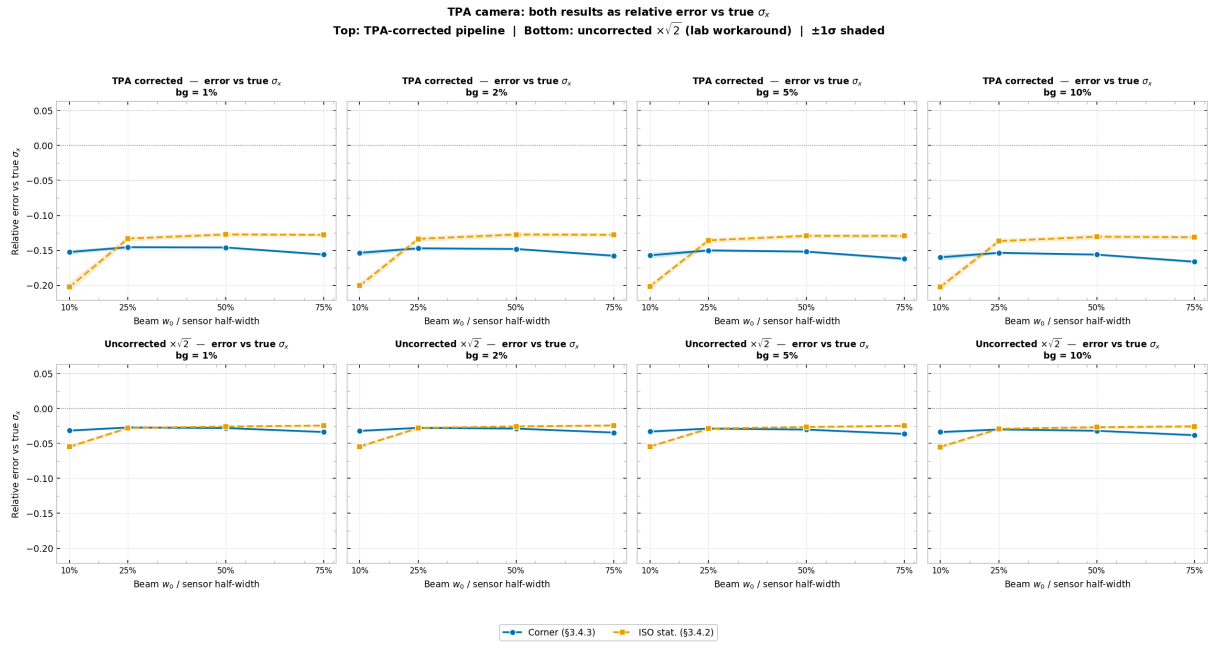


Figure 1: Relative error of both TPA analysis approaches vs true σ_x , as a function of beam/sensor ratio and background fraction. *Top row*: TPA-corrected pipeline (Eq. 2). *Bottom row*: uncorrected $\times\sqrt{2}$ workaround. Both results are expressed as $(\hat{\sigma}_x - \sigma_{x,\text{true}})/\sigma_{x,\text{true}}$. ISO statistical background subtraction. Shaded bands: $\pm 1\sigma$ over 50 noise realisations.

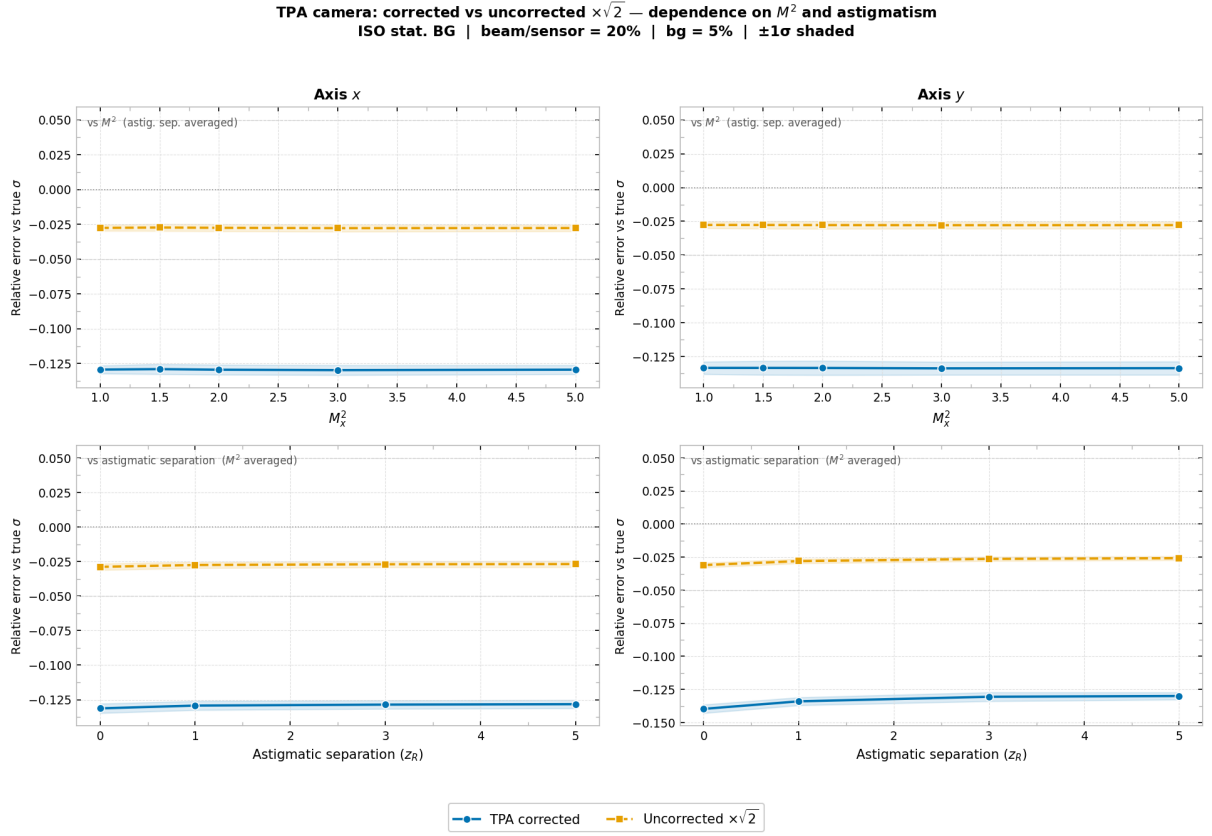


Figure 2: Relative error vs M^2 (top row) and astigmatic separation $\Delta z_0/z_R$ (bottom row) for axis x (left) and axis y (right). Both errors are independent of M^2 and astigmatic separation, confirming that the result holds for any simple astigmatic Gaussian beam regardless of beam quality. Conditions: beam/sensor = 0.20, bg = 5%, ISO statistical background subtraction.

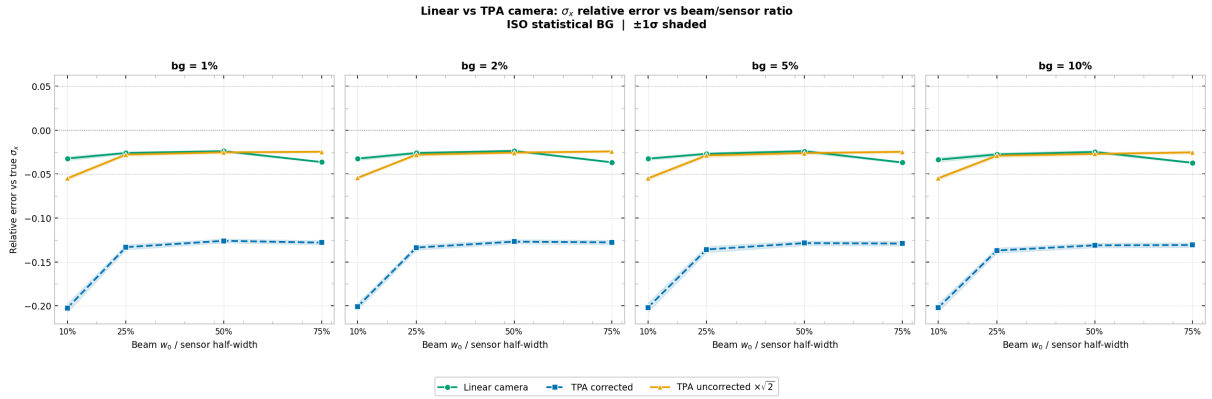


Figure 3: Comparison of σ_x relative error for a linear camera, the TPA-corrected pipeline, and the TPA uncorrected $\times\sqrt{2}$ workaround, as a function of beam/sensor ratio (one panel per background fraction). The linear camera and the workaround track each other closely across all conditions. ISO statistical background subtraction.

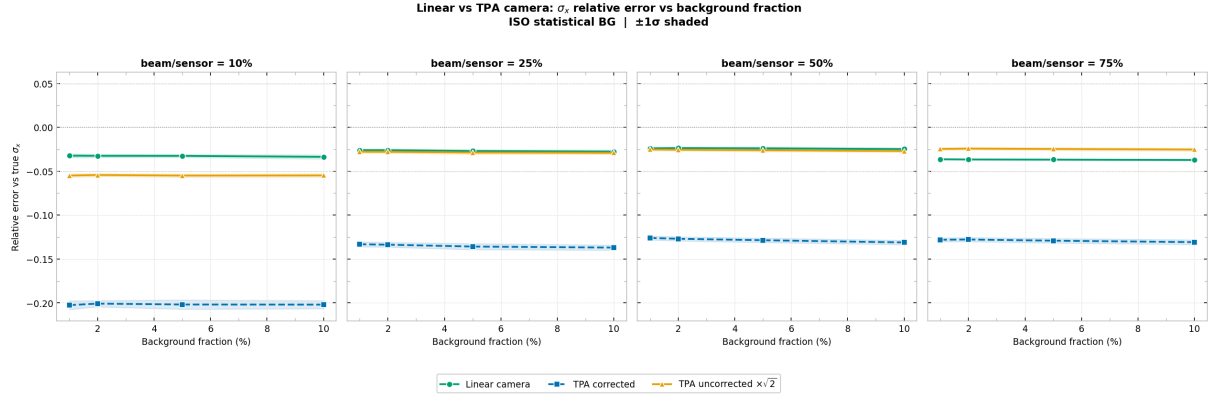


Figure 4: Same comparison as Figure 3 but plotted vs background fraction (one panel per beam/sensor ratio). The TPA corrected pipeline shows a slight dependence on background fraction; the other two approaches are essentially flat.